# Introduction to Economics

Class 14 — Monopolistic competition; problems on Blocks 3–4

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## Plan for today

- Monopolistic competition: many sellers, each a little different
- A coworking space in the short run and the long run
- Why firms advertise
- Practice on the grid: a price taker's costs and profit
- Two curves from a table: a shift, the elasticity; a price cap

## Advertising in Spain: how much a year?

# Revealed in class

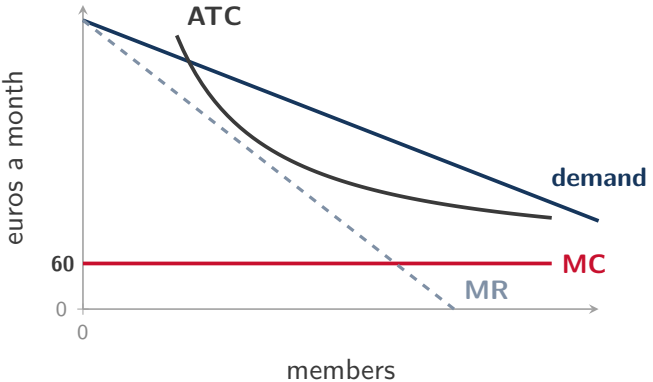
Guess before class: how many euros a year are spent on advertising in Spain, per person living here? 2.60, 26, 260 or 2,600?

# Monopolistic competition: many sellers, each a little different, and anyone can open one

| | Perfect competition (class 12) | Monopolistic competition (today) | Monopoly (class 13) |
|---|---|---|---|
| **Sellers** | many | many | one |
| **Product** | identical | each a little different | no close substitute |
| **Entry** | free | free | blocked |
| **One firm's demand** | flat at the price | slopes down | the whole market's |

Price taker: sells all it wants at the market price (class 12, slide 6) · monopoly: the only seller, no close substitute (class 12, slide 20).

# A coworking space in the short run: how many members, at what price?




60 desks

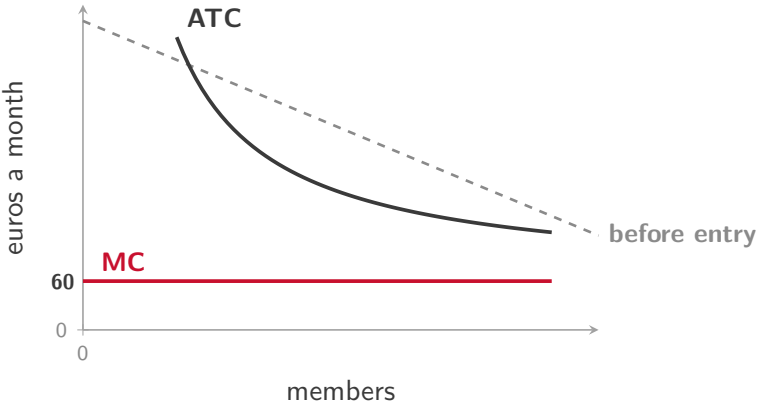
rent, furniture and the owner's time: 3,600 a month

each member costs 60 a month

Marginal revenue (MR): what one more member adds to revenue (class 13, slide 4).

Average total cost (ATC): total cost ÷ members (class 11, slide 10) · MC: marginal cost.

# When new spaces open: what happens to each one's demand?




Profit here is economic profit: every cost counted, the owner's own time included (class 11, slide 6).

# Revealed in class

Compare with class 12, slide 18: where does a price taker end up in the long run?
Reading pp. 270–271.

## Why do firms advertise?

# Revealed in class

Reading pp. 272–276: the critique and the defence of advertising, and of brand names.
What is one more customer worth to a firm whose price is above its marginal cost?

**Monopolistic competition: many sellers, each a little different, free entry.**

**Short run: like a monopoly.**
**Long run: entry until profit is zero.**

**Even then the price stays above MC, and capacity is left unused.**

# Break

Ten-minute break

# A pomegranate farm's costs: the table — four minutes alone

A farm sells crates of pomegranates at the market price of 30 euros a crate; it is a price taker. The table gives its total cost (TC) for 0 to 5 crates a day. [5.5 points; table cells as marked]

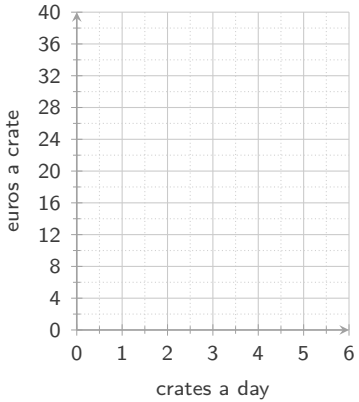
(a) Complete the table: the marginal cost (MC) of each crate (the extra cost of that crate) and the average total cost (ATC) at 2, 3 and 4 crates.

| **Crates a day** | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| **Total cost, TC (euros)** | 24 | 34 | 44 | 54 | 80 | 114 |
| **MC of that crate (euros)** [0.5] | – | | | | | |
| **ATC (euros a crate)** [1] | – | – | | | | – |

TC: total cost · MC: marginal cost, the extra cost of one more crate · ATC: average total cost, TC ÷ crates.

# The farm on the grid: MC, ATC and profit — four minutes alone

The same farm, at the market price of 30 euros a crate. Its costs (euros):




[10.5 points; drawing 4.5; boxes as marked]

| crates a day | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| **TC** | 24 | 34 | 44 | 54 | 80 | 114 |
| **MC** | | | | | | |
| **ATC** | | | | | | |

(b) Draw MC as steps: over each crate, a horizontal line at that crate's MC. Mark the ATC at 2, 3 and 4 crates with dots and join them. Draw the price as a line labelled P.

(c) Best quantity (crates a day) [2] Profit (euros a day) [2]

Shade the profit as a rectangle; write profit inside it.

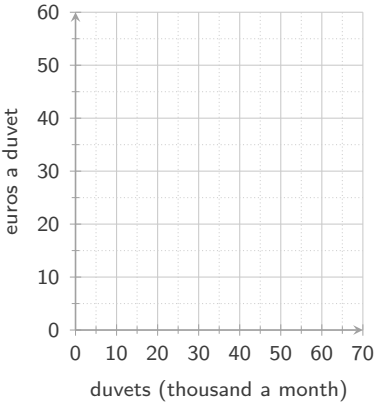
(d) At 12 euros a crate: produce or shut down in the short run? Circle one; the reason in one line. [2]

(Produce) (Shut down)

MC: marginal cost · ATC: average total cost · AVC: average variable cost, variable cost ÷ crates · P: price.

# Duvets: two curves from a table, then a shift — six minutes alone

The market for duvets (quantities in thousand duvets a month). The table gives the quantities demanded and supplied at three prices; both curves are straight lines.




[12 points; drawing 6; boxes as marked]

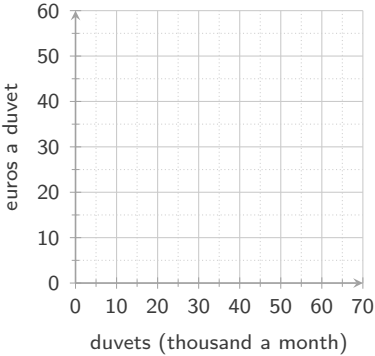
| Price (euros a duvet) | 25 | 35 | 45 |
|---|---|---|---|
| Quantity demanded (thousand a month) | 60 | 40 | 20 |
| Quantity supplied (thousand a month) | 15 | 25 | 35 |

(a) Draw the demand curve and the supply curve through the table's points, label them D and S, and mark the equilibrium $E_1$.

(b) The price of polyester, an input used to make duvets, falls. At every price, sellers now offer 30 thousand more duvets a month than before. Draw the new supply curve, label it $S_2$, and mark the new equilibrium $E_2$.

| | Price (euros) [1.5] | Quantity (thousand) [1.5] |
|---|---|---|
| (a) Before: $E_1$ | | |
| (b) After: $E_2$ | | |

# The duvets: the elasticity from $E_1$ to $E_2$ — two minutes alone




[2 points; working shown]

(c) Demand did not move, so $E_1$ and $E_2$ are two points on the demand curve. Compute the price elasticity of demand between them by the midpoint method, and write it in the box as a positive number (a fraction or a decimal).

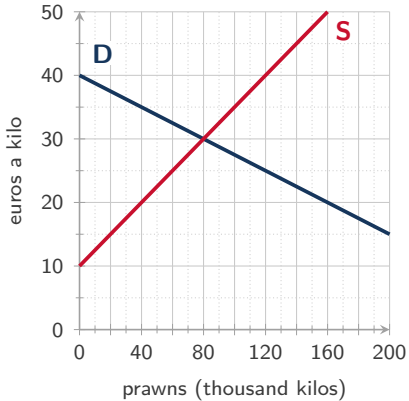
Elasticity (a positive number) [2]

Working:

# Prawns at Christmas: a price cap — five minutes alone

The grid shows a city's market for prawns in Christmas week (quantities in thousand kilos): demand D and supply S. The city sets a maximum price (a price cap) of 20 euros a kilo.

[9 points; drawing 2; boxes as marked]




(a) Draw the cap as a line and label it cap.

(b) The quantity supplied, the quantity demanded and the shortage at the cap (thousand kilos).

Supplied [1.5] Demanded [1.5] Shortage [1]

(c) Only the quantity supplied is sold, and it goes to the buyers who value it most. The deadweight loss (DWL) with the cap, in thousand euros; write DWL inside its area.

DWL [3]

Thousand kilos × euros a kilo = thousand euros. A triangle is ½ × base × height.

## Before Monday 16 November

- Monday 16 November: review for Midterm 1 (Blocks 1–4), the full two hours
- Thursday 19 November: Midterm 1 (Blocks 1–4), in class
- Practise in Sextant: the units the course page lists



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